Microsoft

Azure AI Fundamentals

ASSESSMENT GUIDE

Overview

This document provides information and guidance on how to develop formative and summative assessments for AI900T00A Microsoft Azure AI Fundamentals. It is split into two main sections:

- **Section 1: Presentation Questions –** Guidance for assessing student mastery as they progress through the course. This section includes a set of items for each course presentation that can be used throughout the course to monitor student progress and inform your instruction; the assessment items are a mixture of multiple-choice questions and open-ended questions.
- **Section 2: Capstone Project –** Guidance for using a Capstone project to assess students' ability to apply what they have learned throughout the course to a real-world scenario. This project requires students to create and deliver an Azure Machine Learning talk based on client requirements and includes guidance for both students and the instructor.

This guide is intended to be a reference and starting point for instructors as you plan how to assess your students. As you read through the guide, you may choose to tailor the assessment strategies, including the assessment items and rubric, for your classroom.

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Section 1: Presentation Questions

Introduction

This section includes multiple choice and open-ended questions that are aligned to the course presentations for AI-900. Each presentation includes a sample of five multiple-choice questions and five open-ended questions.

You are free to use the questions as they are currently presented or modify as appropriate for your learners. The questions do not appear in any other course materials and are designed to supplement the formative assessment opportunities that are integrated directly into Microsoft Learn and the Microsoft Official Course (e.g. Knowledge Checks, "Try-It" activities, Exercises, Walkthroughs, Labs). They are also designed to allow for easy integration into an online quiz through Microsoft Forms or through your institution's Learning Management System (LMS). Each set of presentation questions represents indicative coverage for each presentation objective domain.

Assigning the presentation questions to students as an independent activity will enable you to collect data about individual student progress. We recommend that you set aside class time to review answers and address any common student misconceptions.

Overview of multiple-choice questions

The multiple-choice questions require one or more answer responses and will include plausible distractors. These are set at a level slightly lower than the multiple-choice questions in the AI-900 Microsoft Azure AI Fundamentals certification exam.

Overview of open-ended questions

The open-ended questions present further challenges beyond single answer responses and include scenario-based questions. These questions give students the opportunity to demonstrate critical thinking through their responses. As the Capstone project for the course is primarily an Azure Machine Learning pipeline design activity, the open-ended questions at the end of each presentation are a useful approach to help build student confidence and critical thinking to encompass a range of potential solutions.

For their responses, students should be encouraged to explore multiple Azure AI product solutions and adopt a design-first approach before settling on a potential solution. Exploring the official Microsoft Azure AI documentation is a great place for students to research and investigate the different AI products and services that are available.

Presentation 1: Introduction to AI

Multiple choice questions

1. Which of the following human attributes can AI imitate?

    Select all that apply.

    a. Making decisions based on past experiences (correct answer)

    b. Understanding written and spoken language (**correct answer)**

    c. Critical thinking for moral, ethical, and humane behavior

    d. Recognizing abnormal events (correct answer)

2. Which of the following are common AI workloads?

    Select all that apply.

    a. Machine Learning (correct answer)

    b. Blockchain

    c. Computer Vision (correct answer)

    d. Batch processing

3. Which three types of AI workloads could be combined to facilitate a human engaging with an AI agent to submit a suitable profile photo?

    Select all that apply.

    a. **Computer Vision (correct answer)**

    b. Anomaly detection

    c. Conversational AI (correct answer)

    d. **Natural Language Processing (correct answer)**

4. There are challenges and risks associated with developing AI solutions. Which of the following statements is true?

    Select the correct option.

    a. Bias can affect results (correct answer)

    b. An AI algorithm is always correct

    c. Humans are not responsible for AI-driven decisions

d. AI solutions are always more reliable than humans

5. Which of the following is one of the six principles for responsible AI?

   Select the correct option.

   a. Fast performance

   b. Flexible

   c. Inclusiveness (correct answer)

   d. Open source

Open-ended questions

1. A company wants to develop an AI solution for driverless cars. They are considering the ethical implications of such a system and are currently examining the principles of responsible AI. Name one of the principles and how it should be considered in the context of driverless cars.

   Answer: (Should include a short discussion on one principle and how it relates to driverless cars. For example, the 'Reliability & Safety' principle applies to the safeguards that should be built into such a system, such as the AI-driven decisions when a hazard is presented on the road)

2. What important principle for responsible AI ensures that specific groups of people are not disadvantaged. when compared to others, and how might it apply to using a system to apply for a credit card?

   Answer: (Should include a statement on the fairness principle where bias applied to gender, ethnicity and other factors do not negatively impact on the outcome of the application)

3. What is the main developer-centric component of Azure Cognitive Services that makes it quick to integrate in a wide range of software solutions in an agnostic fashion, and how does it save the developer time?

   Answer: (Should include a statement indicating that Cognitive Services is REST API-driven and does not need deep machine learning expertise to integrate in software applications)

4. Which feature of the Azure Machine Learning service allows non-experts to create machine learning models quickly by just submitting data and selecting the desired model? Additionally, briefly describe how this feature works.

   Answer: (Should include a statement that indicates automated machine learning is the feature. It automatically creates algorithms to test against the supplied data and returns a score on how well the algorithm performs)

5. Briefly describe the purpose of anomaly detection and then describe a scenario where it may be used.

    Answer: (Should include a statement that says anomaly detection is used to find unusual changes in data over time. One use of this technology is to detect fraud in banking, for example if a credit card has suddenly been used to purchase several high value items)

Presentation 2: Machine Learning

Multiple choice questions

1. Which of the following statements is true for machine learning?

    Select the correct option.

    a. Machine learning is a technique that we can use to create predictive models that continue to be accurate regardless of new and rapid data changes.

    b. **Machine learning is a technique we can use to create predictive models based on data relationships. (correct answer)**

    c. Machine Learning is a technique we can use to create predictive models that do not rely on data relationships.

    d. Machine Learning is a technique we can use to create non-predictive models based on data relationships.

2. The Azure Machine Learning service has several useful features. Identify these features.

    Select all that apply.

    a. Pipelines (correct answer)

    b. Azure Machine Learning designer (correct answer)

    c. Data and compute management (correct answer)

    d. Natural Language Processing

3. Which of the following settings are required for when creating a new Azure Machine Learning resource?

    Select all that apply.

    a. Azure SQL database name

    b. Virtual network name

    c. Workspace name (correct answer)

    d. Resource group (correct answer)

4. Which of the following are examples of supervised machine learning??

    Select all that apply.

    a. Classification (correct answer)

    b. Regression (correct answer)

c. Clustering

d. Dimensionality Reduction

5. Which of the following is a supervised machine learning type used to predict a numerical value?

Select the correct option.

a. Density Estimation

b. Classification

c. Regression (correct answer)

d. Clustering

Open-ended questions

1. What is a common first task as part of an Azure Machine Learning pipeline created using the designer? Give a brief example of such a task.

Answer: (Should include a statement that states data preparation/ingesting. An example might be cleaning the input data to remove zero values)

2. A business is developing a mobile application to identify plant diseases. They have sourced a large quantity of labelled plant images that exhibit a range of diseases and now want to categorize them using machine learning. Which machine learning type should they use and why?

Answer: (Should include a statement that states that classification should be used to classify the plant diseases by using labels)

3. Medical researchers have a large dataset of labelled brain tumor image scans. They want to group the data by size and shape of the tumors. Which machine learning type should they use and why?

Answer: (Should include a statement that states that clustering should be used based on the features of the tumors including such size and shape)

4. When using Azure Machine Learning designer, what is the Split Data module used for?

Answer: (Should include a statement that says the 'Split Data' module is used to randomly split the dataset in the machine learning pipeline into training and validation subsets, allowing the model to be evaluated using data it wasn't trained with)

5. What is one of the main benefits of using Automated Machine Learning? Include a brief overview of the main steps involved.

Answer: (Should include a statement that states automated machine learning does not require significant machine learning expertise with users able to quickly train and evaluate models. With automated machine learning, the main steps are to supply the service the data, configure the number of iterations for the different features and algorithms, then run and review the results.)

Presentation 3: Computer Vision

Multiple choice questions

1. Which of the following statements is true for the Computer Vision cognitive service on Azure?

   Select the correct option.

   a. Uses pre-trained models to analyze images. (correct answer)

   b. You need to train your own model to analyze images.

   c. You can train your own model or use a pre-trained model to analyze images.

   d. It is not accessible through a REST API.

2. Which of the following would be used to indicate the location of an item within an image?

   Select all that apply.

   a. Image classification

   b. Face detection and recognition (correct answer)

   c. Object detection (correct answer)

   d. Optical character recognition

3. Which two parts of information does your application need to consume Computer Vision cognitive services?

   Select all that apply.

   a. REST endpoint (correct answer)

   b. Password

   c. Subscription key

   d. Authentication key (correct answer)

4. You want to train your own image classification model for different car models. Which service should you use?

   Select the correct option.

   a. Form Recognizer

   b. Computer Vision

   c. Custom Vision (correct answer)

   d. Video Indexer

5. Which of the following facial analysis feature does the Computer Vision service offer?

Select all that apply.

a. Emotion recognition

b. Face verification

c. Age (correct answer)

d. Gender (correct answer)

Open-ended questions

1. A company wants to implement the Azure Computer Vision service as part of a software application. What are the two main development approaches they can use to integrate the Computer Vision service in their application, and what is the main difference between them?

Answer: (Should include a statement that the main approaches are the direct REST API (via HTTP) approach and a language-specific client library approach (e.g. C# or python). The main difference is the direct REST API approach is generally language agnostic while the client libraries are designed to be used with a specific programming language)

2. When training a model for a set of images, what is the difference between image classification and object detection?

Answer: (Should include a statement that discusses image classification in the sense that it applies one or more identifying labels to an image, while object detection returns the location/coordinates within the image where the label has been applied)

3. A company needs to develop an application that visually identifies employees as part of their security protocols for accessing secure areas. What cognitive service should they use and why?

Answer: (Should include a statement that states the Face service is the most suitable for this task as it can identify and verify that two faces belong to the same person with a good degree of confidence)

4. The Form Recognizer service is made up of a small collection of sub-services. What are the sub-services, and which one would be used by a school to automatically process an attendance monitoring form they use?

Answer: (Should include a statement that states the sub-services are Custom models, prebuilt models, and Layout API. The school would use the 'Custom models' service to extract the key/value data values from the attendance forms)

5. A university wants to use the Azure Computer Vision API to extract text from handwritten assessments to aid the marking process. The handwritten assessment documents are first scanned as PDF files. Which capability of the Computer Vision API would be suitable for this task and why?

Answer: (Should include a statement that states the Computer Vision Read API is suitable to extract text from the scanned PDF files. The Read API can extract text from images and PDF files and can detect the handwritten text in the assessment documents)

Presentation 4: Natural Language Processing

Multiple choice questions

1. Which of the following tasks is associated with Natural Language Processing (NLP)?

Select all that apply.

a. Sentiment analysis (correct answer)

b. Text moderation

c. Semantic language modelling (correct answer)

d. Speech recognition and synthesis (correct answer)

2. A company wants to understand how its brand is perceived on social media platforms. Which feature of the Language API would you use to quantitively measure how positive or negative user-generated content is?

Select the correct option.

a. Key phrase extraction

b. Language detection

c. Sentiment Analysis (correct answer)

d. Named entity recognition

3. Which of the following are components of Conversational Language Understanding?

Select all that apply.

a. Intent (correct answer)

b. Sentiment

c. Utterances (correct answer)

d. Entities (correct answer)

4. A company wants to use a Speech service to capture real-time audio from interview discussions and convert to text. Which API should they use?

Select the correct option.

a. Text-to-Speech

b. Speech-to-Text (correct answer)

c. Translator

d. Batch transcription

5. Which feature would you use to improve the accuracy and quality of a Custom Question Answering knowledge base?

Select the correct option.

a. Face service

b. Active learning suggestions (correct answer)

c. Key phrase extraction

d. Ink Recognizer

6. Which of the following two services can be used to create a chat bot for a knowledge base?

Select all that apply.

a. Optical Character Recognition

b. Azure Bot Service (correct answer)

c. Language (correct answer)

d. Speech

7. Which of the following statements best describes conversational AI?

Select all that apply.

a. A solution that enables text translation

b. A solution that enables text-to-speech

c. A solution that enables a dialog between AI agents

d. A solution that enables a dialog between an AI agent and a human (correct answer)

Open-ended questions

1. A company wants to develop a smart speaker that responds to voice commands. What service could they use to respond to user commands to control the speaker, and how would it work?

Answer: (Should include a statement that says they can use the Conversational Language Understanding feature of the Language Service. The service can be used to train a language model involving the components of Utterances, Entities, and an Intent to interpret natural language commands)

2. A company wants to analyze hotel reviews so it can understand customer perceptions of hotels they have stayed at. What service could they use, and how would it work?

> **Answer: (Should include a statement that says they could use the Text Analytics feature of the Language Service to extract sentiment (a numerical value for positive and negative) and key phrases such as 'fantastic evening meal' from the submitted review text)**

3. When using the sentiment analysis feature on a document, a score is returned. What is the range of this score, and what does it mean in terms of the contents of the document?

> **Answer: (Should include a statement that says a sentiment score of between 0 and 1 is returned to two decimal places. A score of 0.5 is neutral while less than this indicates negative sentiment and more indicates positive sentiment, dependent on the content in the document)**

4. You are developing a mobile application that will have the capability to take audio input from a microphone and transcribe it in real-time on the mobile display. What service should you use and why?

> **Answer: (Should include a statement that says the Speech service can be used to do the task. It can translate audio sources in different languages to text, which can then be displayed on the screen)**

5. There are two main tasks involved in creating a Conversational Language Understanding application. Name each of them and their purpose.

> **Answer: (Should include a statement that says the tasks are 'authoring' and 'prediction'. The authoring task involves defining the entities, intents, and utterances to be used, referred to as authoring the model. The prediction task is publishing the authored model for intent and entity prediction, which is based on the user input)**

6. What service would you use to create an automated FAQ and why?

> **Answer: (Should include a statement that says the Custom Question Answering feature of the Language service can be used. Custom Question Answering can be used to create a knowledge base of answers that can respond with the same answer when different users submit the same or similar questions)**

7. When creating a Custom Question Answering knowledge base, what is meant by the term *alternative phrasing* for questions?

> **Answer: (Should include a statement that says alternative phrasing is based on anticipating the different ways a question can be asked and responding with the same answer. This is essentially consolidating questions with the same meaning)**

Section 2: Capstone Project

Overview

This guide features a single, end-of-course Capstone project in which each student creates and delivers an Azure Machine Learning pipeline design solution presentation based on client requirements. The project draws upon the students' knowledge and understanding gained from all presentations in the course and presents an opportunity to demonstrate their intellectual curiosity around the design and presentation of a scenario-based Azure Machine Learning pipeline solution. It is designed to be flexible enough to be used in a variety of contexts for students of various technical levels.

The Capstone Project Student Guide can be copied and redistributed to students as standalone content, as it provides an example of a baseline project that could be tailored to fit the needs of various learners. You may choose to modify some of the content dependent on which parts of the Capstone are to be assessed and whether a real-world client or pre-defined scenario is used.

The Capstone Project Educator Guide provides additional guidance for modifications that can be made to the Capstone project, as well as comprehensive information on best practices in the preparation and development of the project.

The Capstone Project Rubric provides additional information for students and teachers in using the project as a summative assessment. Students may use this document to guide them as they complete the activities, ensuring that their work reflects what they have learned in the course.

Capstone Project - Educator Guide

Overview and Objectives

This is the final activity for the course. The primary objective is to create a presentation that summarizes and justifies an industry client's needs and goals for an Azure AI solution, along with a diagrammatic representation of the proposed solution. The presentation should also document the estimated service costs and state the benefits to the client. This is an opportunity for students to reflect on what they have learned throughout the course, both about Microsoft Azure AI services and the client. There is a specific focus on the Azure Machine Learning services and tools for the Capstone, which are part of the Azure AI platform and covered in presentation 2 of the course. The reasoning behind the Machine Learning focus is that the knowledge and skills gained from this area are the foundations of the course. If you wish to modify the Capstone project by extending the supplied pre-defined scenario into other areas of the course such as Computer Vision or Natural Language Processing, then please read the **Capstone Modification Suggestions** section of this guide.

Capstone Modification Options

As you plan your course, you may choose to tailor the Capstone project to the unique needs of your students. This may be in response to particular learning goals or logistical constraints. Depending on the student cohort and the focus of their study program, you may want to consider changes to:

- The parts of the Capstone students will be assessed on.
- The use of either a real-world client or pre-defined client scenario.
- The rubric criteria

The Capstone can be completed individually or in groups and is modular to suit different student cohorts. You can select which parts of the Capstone project to deploy for assessment purposes.

To align with industry expectations and real-world application of cloud-driven AI skills, the Capstone is primarily designed as a scenario-based project. In the baseline version of the project, students interact with a real-world industry client, allowing them to experience the requirements gathering phase and carry out research around configuring suitable AI services and associated components. If this is not possible, you can develop a pre-defined industry scenario with which students can make assumptions based upon the scenario content to develop an informed set of requirements. You can also ask students to research an existing company and develop a set of recommendations based on publicly available information.

The rubric criteria can easily be modified to reflect the priorities with your learners. You may choose to add or remove rows from the rubric or change the details of the scoring criteria.

Capstone Parts

The Capstone is split into three parts that offer some flexibility to adapt to student needs and their program of study:

- Part 1 – Present the requirements for the real-world client or pre-defined scenario with a proposal that frames how Azure Machine Learning, specifically the Azure Machine Learning designer tool, offers the capability to design an Azure Machine Learning pipeline solution to meet the client needs and goals. In addition, students should propose a bulleted list of modules and the selected algorithm the pipeline

solution will include. The reference guide for the modules and algorithm(s) that are part of the Azure Machine Learning designer tool are here.
- Part 2 – Design an Azure Machine Learning pipeline diagram based on the available module and algorithms that are part of the Azure Machine Learning designer tool. The diagram should show the pipeline workflow in a linear fashion with modules and algorithm(s) clearly labeled.
- Part 3 – Present the pipeline solution with a discussion on the individual modules and selected algorithm(s) that are part of the pipeline diagram.

It may be more appropriate to assess students on less-technical programs against **Part 1** only. Students on more technical programs may be assessed across **Parts 1,2** and **3**. This is a judgement that you, as the instructor, are best equipped to make.

Part 1 – Present how Azure Machine Learning, specifically the Azure Machine Learning designer tool, offers the capability to design a solution

Depending on how you deliver the course, students can use class sessions or independent study time to complete their Capstone work. This can also be facilitated through remote drop-in/support sessions.

Remind students to review the Capstone rubric so they understand the required tasks for Part 1 as listed below:

- A summary description of the client's current situation, needs, and goals.
- Proposal of how Azure Machine Learning, specifically the Azure Machine Learning designer tool, provides capabilities to design an Azure Machine Learning pipeline solution to meet the client goals.
- A bulleted list of the modules and selected algorithm(s) proposed for use as part of the pipeline solution. Only the modules and algorithms that are part of the Azure Machine Learning designer tool can be used.
- Expected costs – use the online Azure Pricing Calculator for this.
- Benefits to their clients.

Part 2 – Design an Azure Machine Learning pipeline diagram

This part of the Capstone project is designed specifically for students in more technical programs. It involves an understanding of Azure Machine Learning pipelines and the modular components they are comprised of. The diagram can be created in Microsoft PowerPoint or Microsoft Visio. Alternatively, there are several free to use web-based diagramming tools that are suitable for the task. An example diagram is presented in Appendix B for the pre-defined client scenario for an Azure Machine Learning pipeline solution. The company requires the solution for predicting the energy consumption of houses that are for sale, based on specific attributes.

Remind students to review the Capstone rubric so they understand the required tasks for Part 2 as listed below:

- Visualize an Azure Machine Learning pipeline based on the available algorithms and modules that are part of the Azure Machine Learning designer tool.
- Pipeline should visualize the machine learning workflow in a linear fashion with each module labelled appropriately.
- Identifies the selected algorithm(s).

Part 3 – Present the Pipeline solution with a discussion on the workflow and chosen algorithm

Students can deliver their presentations either in-class or remotely. In terms of timings, approximately 10 minutes for the presentation followed by 5 minutes for questions should suffice. Consider the following options to decide the best format for your students to present their work to an audience:

- **Whole-class presentations:** Students share their presentations with the whole class, taking turns and following any time limits – this is only suitable for small class sizes.
- **Small group presentations:** Divide the class into small groups. Allow each student a set amount of time to present to the other students in the group.
- **Invite the clients:** If possible, inviting the clients to the presentations will provide the students with further feedback opportunities.

Students should be encouraged to discuss and justify why they selected the pipeline modules and algorithm(s) to meet the client's needs. You may want to ask students to respond to some or all of these reflection questions:

- Why did you select the modules used in the pipeline?
- Why did you select the algorithm(s) used in the pipeline over others?
- What benefits might the client customers expect?
- What was the most challenging part of the project?
- If you had more time what improvements would you make?

Client Scenarios

In the baseline project as outlined in the student guide, students are assumed to have access to a real-world client. However, you may also choose to have students research information about a client or provide them with a pre-defined client scenario.

Real-world client

The benefits when using a real-world client are clear: it provides a measure of authenticity for the students and also aligns with industry expectations. Generally, the instructor would be responsible for sourcing one or more clients with whom the students can have scheduled time with to understand their current business needs and goals. Ideally a session would be arranged with the client in the first instance during which they can communicate their needs and goals to all students, followed by a Q & A session.

The instructor should guide the client to communicate the following information for the students to ensure students have all the information they need to begin their preparatory work:

- Company type
- Products
- Customer expectations
- Current technology resources
- Projected needs and goals

Researched client

If direct access to real-world clients is not available, you may choose to have students research a client using publicly available information. This allows for a sense of authenticity without the logistical challenges of managing client interactions. In this case, you may have students choose from a list of appropriate clients that you have pre-selected or allow each student to submit their own potential client for approval. Be sure to confirm that students can find the same information listed in the previous section from publicly available information when creating a list of appropriate clients and/or approving clients submitted by students.

Pre-defined client

If a client is not be available or if there are time constraints to deliver the course, you can also choose to give students a pre-defined client scenario. See below for a scenario that can be used as a pre-defined client. You are free to modify the scenario or develop one of your own.

The client company wants to develop an energy consumption predictor feature as part of a mobile application for customers who are in the market to buy a house. Customers will be able to search for a house type with attributes such as floor space, number of rooms, number of levels, construction type, and year it was built. Based on the information entered by the customer, the application would return a response that lists house matches with a prediction of the yearly energy consumption cost.

In order to offer this functionality as part of their mobile application, the company has a large dataset that contains data on thousands of houses. The house data information includes floor space in square feet, number of rooms, number of levels, number of bathrooms, window type, garden, zip code, construction type, year it was built, last sold price, and the yearly energy consumption in kWh and cost. The company wants to use the dataset to train a model using Azure Machine Learning, specifically by developing a pipeline using the Azure Machine Learning designer tool and publish the pipeline so that it can be used by their mobile application. Your role is to design the pipeline through a diagram, not implement it, using the available modules and algorithms as part of the Azure Machine Learning designer tool.

The company believes this will give their mobile application a market advantage over their competitor's offerings and will be a useful tool for potential buyers looking for a more economical and environmentally friendly house.

The following information can be derived from the client scenario above, which is similar to the basic information that can be derived from a real-world client:

- **Company type** – Specializes in offering services for house buyers.
- **Their products** – Mobile applications for customers to search for houses for sale that meet their needs.
- **Their customers' expectations** – Customers expect to be able to save money by finding houses for sale that are more economical to run and environmentally friendly.
- **Their current situation and how they do things now** – Currently, they do not offer an energy consumption predictor value for houses that are returned through customer searches on their mobile application
- **Their projected needs and goals** – The company would like to market themselves more as an environmentally aware business and believes their mobile application would have the advantage over their competitor's offerings as customers would be able to search for, and find homes that offer more value in terms of overall costs and environmental impact.

Capstone Modification Suggestions

If you prefer to offer your students a Capstone project that is focused on other areas of the course, then you may wish to consider the below suggestions. To modify the Capstone using the suggestions you will need to create the required content, however, you are free to modify any of the materials presented in this guide to support it.

- **Conversational AI**: Extend the pre-defined client scenario by designing an online chatbot using the Azure Bot Service. The chatbot should facilitate conversation flows around users asking for environmental impact information on houses for sale.
- **Computer Vision**: Develop a new Capstone project for road traffic monitoring that can identify images of the various types of vehicles using the Custom Vision Service.

- **Language Understanding**: Develop a new Capstone project that uses Conversational Language Understanding to develop a conversational interface around IoT devices, specifically by using voice commands with a smart home hub. The hub should have the capability to control other devices in the home such as connected lights, speakers, heating, televisions, fridges, and home security devices such as cameras

Preparing and Supporting Students

We recommend that you introduce the Capstone after Presentation 1 of the course to allow students to start thinking about how they can apply what they are learning each week to the project. The end-of-presentation questions, particularly the questions with an open-ended format, lend themselves well to preparing your students to think independently about the processes to design a suitable Azure Machine Learning pipeline for the project. See the introduction to the **Presentation Questions** section for suggestions of how to encourage student engagement to help scaffold the process of designing a solution for the Capstone project.

If students have questions or are facing problems while completing the tasks, direct them to the following problem-solving strategies before asking you for help:

- Review the content in the Student Capstone Guide.
- Research the official Microsoft Azure AI Documentation.
- Ask peers for help.
- Review their client notes/pre-defined scenario document.

You may consider adding extra sessions as needed for the Capstone preparation and presentation delivery. This will give students dedicated collaborative time to work in groups and allow you to informally monitor their progress as they move through the various tasks.

Capstone Project - Student Guide

Overview

In this Capstone, you'll create a presentation that summarizes your client's current situation, needs, and goals; describes the services and tools you'd recommend for your client; documents the expected costs; outlines a machine learning pipeline diagram; and states the benefits for your client. This is an opportunity for you to reflect on what you have learned throughout the course about Microsoft Azure AI and your client that requires a solution.

Preparing for the Capstone

To best prepare for the Capstone, consider the following:

- **Start preparing early**. Read this Capstone guide early in the course.
- **Meet with your client and take notes**. Take and keep thorough notes about your client and remember: it's better to get too much information from your client than not enough.
- **Think about your client in each presentation**. As you learn new concepts in each presentation, consider how they might apply to your client.
- **Work on the presentation early and complete the end-of-presentation questions**. Consider building your presentation throughout the course. The end-of-presentation questions are designed to help support your understanding of the Capstone tasks, so think about your project as your complete them.
- **Refer to the rubric as you work**. Use the Capstone rubric document to ensure you fully understand the requirements for the tasks presented below.

Capstone Tasks

Task 1 - Produce a list of appropriate cloud services for the client

In the first phase of this project, you will be working to understand your client's needs and using what you know about Azure Machine Learning services and tools to design a solution. By the end of this task, you should have a clear idea of what your client's needs are and how your solution meets those needs. You will be using this information to create a presentation, so be sure that you have all the following documented:

- A summary description of the client's current situation, needs, and goals
- Proposal of how Azure Machine Learning, specifically the Azure Machine Learning designer tool, provides capabilities to design an Azure Machine Learning pipeline solution to meet the client goals.
- A simple bulleted list of the modules and the selected algorithm(s) you propose to use as part of your pipeline solution design. Technical background on each of the machine learning algorithms and modules can be found in the Algorithm and Module Reference.
- Expected costs (use the online Azure pricing calculator for this)
- Benefits to the client.

Task 2 – Design an Azure Machine Learning pipeline diagram

Use the proposed list of modules and algorithm(s) identified in Task 1 and integrate them as an Azure Machine Learning pipeline workflow diagram. The diagram can be created in Microsoft PowerPoint or Microsoft Visio. Alternatively, there are several free to use web-based diagramming tools that are suitable for the task.

Be sure to include the following in your diagram:

- Diagram of an Azure Machine Learning pipeline based on the available algorithms and modules that are part of the Azure Machine Learning designer tool.
- Visualization of the machine learning workflow in a linear fashion with each module labelled appropriately.
- Identification of the selected algorithm(s).

Task 3 – Present the cloud solution

In this phase, you will deliver a ten-minute presentation that describes the needs analysis, recommendations, and explanations that you documented earlier in the project. After the presentation, others will have an opportunity to ask you questions about your process and recommendations. Make sure you include clear evidence to support the decisions that you have made, based on the information that you gathered about your client, and that your presentation is organized and detailed enough for your audience to have a good understanding of the benefits of your recommendations.

Support and Resources

If you have questions or are facing problems while completing the tasks, use the following problem-solving strategies before asking your instructor for help:

- Review the content in this Student Capstone Guide.
- Research the official Microsoft Azure AI Documentation.
- Ask peers for help.
- Review your client notes.

Post-Project Reflection Questions

1. What did you learn in creating and delivering this Capstone presentation that might be helpful to you in the future?

2. What challenges did you experience? How did you work to overcome them?

3. What components of your presentation went well? What are some things in your presentation that you could improve?

4. Add other comments you would like to make about this project or your work.

Appendix A – Capstone Rubric

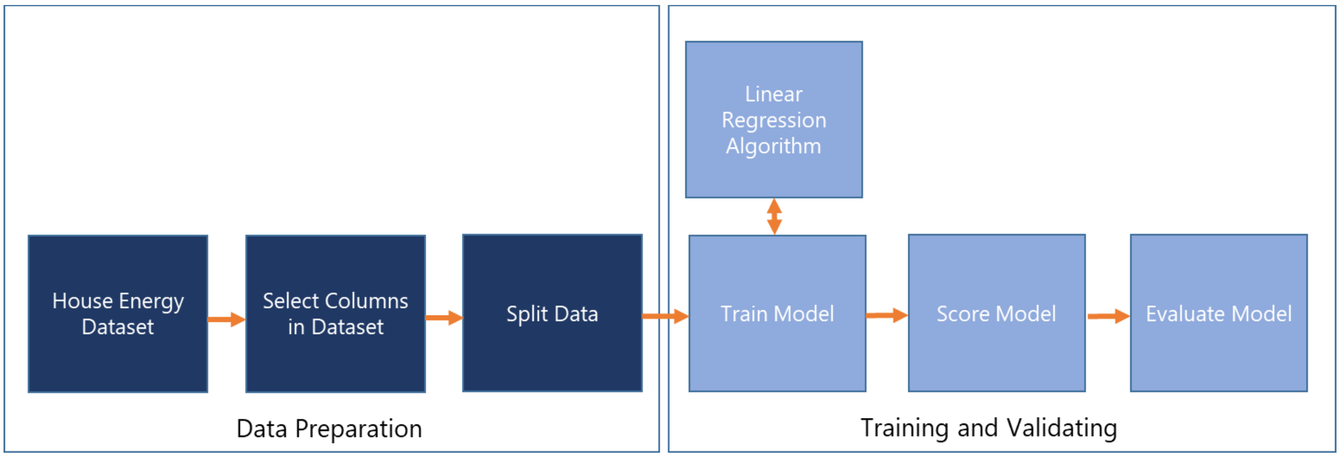
Item	Approaches standard	Meets standard	Exceeds standard (Includes items in Meets Standard)	Summarize where & how you included this item
Task 1				
Description of client's current cloud situation, goals, and needs	Description of the client's situation, goals, and needs aren't clear and relevant.	Description of the client's situation, goals, and needs are clear and relevant to the business.	Description includes direct quotations from the client that provide details about their situation, goals, and needs and that are relevant to the business.	
Solution proposal frames how Azure Machine Learning can support the client's goals	Not all proposals / recommendations fit the client or demonstrate an understanding of appropriate Azure Machine Learning products and services.	Proposals and recommendations fit clearly with an understanding of the client. Services from the course are considered as potential solutions. Reasoning for recommendations demonstrates an understanding of Azure Machine Learning concepts.	Proposals and recommendations at times evidence critical thinking and explain products and services that weren't chosen and a rationale as to why.	
Description of the costs expected for the client	Costs aren't clearly explained or lack detail, or the costs don't include support or a service-level agreement (SLA).	Costs are clearly explained in a detailed way that's easy to understand. All costs are considered, including support and SLAs.	All costs include supporting information that relates cost to the client's specific needs.	
Description of the benefits for the client	Client benefits aren't clearly explained or lack detail, or a monthly cost calculation isn't included.	Client benefits are clearly explained in a detailed way that's easy to understand and includes a monthly cost calculation.	Benefits include supporting information that relates to the client's current situation and	

			potential needs in the future.	
		Task 2		
Visualize an Azure Machine Learning pipeline diagram using the list of proposed modules from task 1	Diagram conveys a basic, but somewhat limited pipeline solution for the client's needs. Some module types may not be the best fit or are missing.	The diagram presents a robust infrastructure solution that should meet most of the client requirements. An appropriate range of module types are evidenced and presented in a linear pipeline workflow.	The diagram presents a well-considered pipeline workflow that should fully support the client's needs. A wide range of module types are considered to a good standard and presented in a linear workflow.	
An appropriate pipeline algorithm(s) is selected and positioned correctly in the pipeline workflow	The algorithm is not a good fit or is missing from the pipeline.	The selected algorithm is suitable for the solution and positioned as part of the pipeline.	The selected algorithm fully meets the client needs and is positioned correctly in the pipeline workflow.	
		Task 3		
Presentation aids	Not all materials are organized or easy to understand. Not all visual and/or audio elements help audience understanding, some might distract.	Materials are organized and clear. Visual or audio elements help audience understanding.	Materials are interesting, easy to understand, and include at least one way to gather audience responses beyond just asking if there are any questions.	
Delivery of presentation	Presenter isn't prepared or doesn't engage with the audience.	Presenter is prepared and engaged with the presentation and the audience. Communication of ideas is mostly clear and effective.	Presenter communicates beyond just reading the words on the presentation materials. Communication of ideas is	

			consistently clear and effective.	

Appendix B – Example Capstone Machine Learning Pipeline Diagram

The diagram presented below is an indicative example of what a student could produce as a basic pipeline diagram to address the pre-defined client scenario in the Client Scenarios section. More advanced modules can be added, for example adding more data transformation modules to clean and normalize the data before training the model.



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